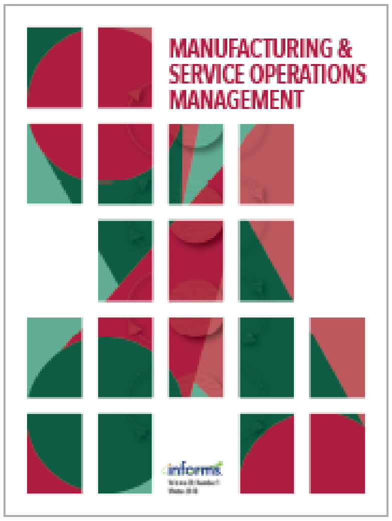

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Strategic Cross-Subsidization in Healthcare Capitation Programs: Evidence from Medicare Advantage

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Abstract. ***Problem definition*:** This study identifies a resource misallocation problem in Medicare Advantage (MA), the United States' largest healthcare capitation program, which may result in discrepancies between patients' health status and the healthcare resources allocated to them. ***Methodology/results*:** Utilizing a large commercial insurance database with claims from more than 2 million MA enrollees, this research investigates the allocation of MA capitation payments. By exploiting an exogenous policy shock on MA capitation payments through a difference-in-difference design, we find empirical evidence of an illegal practice known as "cross-subsidization." This practice involves MA health plans strategically reallocating portions of the capitation payments intended for one group of patients to spend on another group of patients. Additionally, we show that this cross-subsidization practice is associated with the risk selection problem in MA, where low-risk patients are more likely to enroll in MA compared with high-risk patients. ***Managerial implications*:** This research unveils a previously undocumented healthcare resource misallocation problem, that is, strategic cross-subsidization. This practice is explicitly prohibited by law in the United States due to its heightened effect on the undesired risk selection within capitation programs, where health plans cherry-pick profitable enrollees through strategic benefit designs. Our study has direct practical implications as it underscores the need for greater transparency in MA claims data to enable the Centers for Medicare & Medicaid Services to more effectively administer the MA program.

Supplemental Material: The online appendices are available at https://doi.org/10.1287/msom.2023.0637.

Keywords: healthcare markets • capitation payment models • risk selection • difference in differences

1. Introduction

Capitation has been highlighted as one of the most promising healthcare payment models and is increasingly adopted by healthcare payers around the world (James and Poulsen 2016, Lai 2022). In particular, over the last decade, the Centers for Medicare & Medicaid Services (CMS), the largest healthcare payer in the United States, has regarded capitation as one of its targeted payment models (Rajkumar et al. 2014, Centers for Medicare & Medicaid Services 2015). CMS oversees the largest healthcare insurance program in the United States, that is, Medicare. The current Medicare program consists of two components: (1) traditional Medicare (TM), which operates under traditional volume-based payment models, and (2) Medicare Advantage (MA), which is a capitation program. Over the past 10 years, the size of MA has more than doubled, from 13 million enrollees in 2012 to 28 million in 2022, corresponding to 48% of the total Medicare population (Freed et al. 2022).

The allocation of capitation payments and healthcare resources in MA is determined by MA health plans instead of CMS. By design, CMS pays MA health plans a risk-adjusted capitation rate for each Medicare beneficiary enrolled and authorizes the health plans to use the capitation payments to manage the overall health of their enrollees. Therefore, MA health plans have flexibility in operations management decisions, such as which providers to contract with or at what level to offer various direct or nondirect healthcare services. For example, some MA health plans offer free or discounted gym memberships, which are not offered in TM (Cooper and Trivedi 2012). Furthermore, as MA health plans are not required to submit their claims data to CMS, the operations of these MA health plans remain a black box to the public. This lack of transparency in MA health plans' operations, as argued in a recent paper published by the flagship journal of the American Medical Association, raises questions about "the relative value Medicare

Advantage provides to beneficiaries" (Brennan et al. 2018).

This paper aims to open the black box of MA health plans' operations by empirically studying their allocation of capitation payments among Medicare beneficiaries. Leveraging a large commercial insurance database containing claims from more than 2 million MA enrollees, our goal is to delineate the alignment, or lack thereof, between the intended and actual healthcare resource allocations in MA health plans. This investigation was partly inspired by a statement made by a representative of the American Medical Association (AMA):

> "If your health is good, maybe these plans represent value for some patients, like providing gym memberships … . But that can change in the blink of an eye- … with a stroke or an accident or some acute medical condition and they need a rehabilitation stay. Then, services are restricted so much that they can't recover adequately from the stroke … " (MedPage Today 2020).

This statement is consistent with an extensive body of medical literature, highlighting that MA health plans tend to overspend on low-risk patients and underspend on high-risk patients (Cooper and Trivedi 2012, Rahman et al. 2015, Hung et al. 2016, Meyers et al. 2018).

In this paper, we explore a potential mechanism driving the spending disparities between low-risk and high-risk patients within MA health plans. Specifically, we posit that a significant factor contributing to this discrepancy is the strategic cross-subsidization behavior of MA health plans, where the health plans deliberately redistribute a portion of the capitation payment originally designated for one group of patients to spend on another group of patients. This hypothesis reflects the possibility that MA health plans, in order to offer additional benefits such as fitness memberships to some members, might need to curtail certain benefits from others to "free up funds." Moreover, when faced with a tradeoff between investing in different benefits, MA health plans tend to opt for "lower-cost benefits that may broadly attract more enrollees" at the expense of "benefits targeted to high-cost patients" (Skopec et al. 2019b, p. 7).

To empirically identify the strategic cross-subsidization mechanism, we leverage a payment shock from the Affordable Care Act (ACA). This shock incrementally reduced MA capitation payments over time, imposing deeper cuts for high-risk patients compared with their low-risk counterparts. In the absence of cross-subsidization, one would expect this shock to result in deeper spending cuts for high-risk patients. However, through a difference-in-differences (DID) design, we demonstrate that this shock led to deeper spending cuts for low-risk patients instead. This finding suggests that spending patterns in MA are not entirely driven by payment patterns, supporting our cross-subsidization hypothesis. Importantly, this result challenges the prevailing explanation in the literature that observed spending trends might directly stem from payment inaccuracies, where the MA risk adjustment formula tends to overpay low-risk patients and underpay high-risk patients (Brown et al. 2014, Glazer and McGuire 2000). Our identification strategy (detailed in Section 6.3.1) enables us to differentiate the proposed cross-subsidization hypothesis from the competing argument of payment inaccuracy, demonstrating that the observed spending trends within MA, that is, overspending on low-risk patients and underspending on high-risk patients, cannot be solely attributed to overpayment for the former group and underpayment for the latter.

In addition, this paper explores the implications of strategic cross-subsidization, investigating its possible link to the risk selection problem in MA. Although direct patient selection is prohibited in MA, MA health plans may subtly skew the provision of certain services to attract healthier patients and under-provide other services to deter sicker patients. In fact, despite the rapid increase in overall enrollment in MA, a growing number of sicker Medicare beneficiaries are observed to switch from MA to TM (Morrisey et al. 2013; Jacobson et al. 2015, 2019a; Rahman et al. 2015; Li et al. 2018; Meyers et al. 2018, 2019). This patient self-selection phenomenon, widely referred to as risk selection, is considered a major obstacle inhibiting the smooth functioning of individual health insurance markets such as MA (Geruso and Layton 2017). We propose that MA plans may engage in risk selection through strategic cross-subsidization, redistributing funds from high-risk to low-risk patients, thus discouraging the enrollment of the former and encouraging the latter. This form of cross-subsidization differs from that in group health insurance markets. In group health insurance, a plan covers a predefined group of people, such as all employees within a company, and must cross-subsidize unprofitable patients with funds from profitable ones to pool revenue effectively. In contrast, MA health plans have the flexibility to use cross-subsidization to influence the risk level of patient enrollment, rather than to solely focus on revenue pooling within a given patient population. To empirically establish the link between strategic cross-subsidization and risk selection, we evaluate the impact of the ACA cuts in MA capitation payments on risk selection. Specifically, if MA plans conduct risk selection through strategic cross-subsidization, the ACA cuts should alleviate risk selection within MA, as fewer capitation payments can be cross-subsidized from the high-risk to the low-risk after the ACA cuts. Conversely, if MA plans do not employ this strategy, the ACA cuts could exacerbate risk selection within MA, as plans have stronger incentives to discourage high risk patient enrollment due to reduced capitation payments. Our second hypothesis scrutinizes the impact of the ACA policy shock on risk selection within MA and finds additional support for our proposed cross-subsidization mechanism.

Overall, our paper makes two important contributions to healthcare operations management literature with significant implications for health policy and practice. First, we provide, to our knowledge, the first empirical evidence of a previously unknown resource misallocation problem in MA and identify strategic cross-subsidization as the underlying mechanism. Although previous operations management literature has examined various resource misallocation problems in healthcare operations, these tend to focus on local settings such as certain emergency departments, nursing homes, or hospitals (Kc and Terwiesch 2011, Lu and Lu 2017, Ibanez et al. 2018). Our work investigates a broad but unrecognized resource misallocation problem in health plans. Second, our paper underscores a contentious practice in MA, thereby drawing attention to the necessity for CMS to demand greater transparency in MA claims data. The U.S. Code of Federal Regulations explicitly states that "Payments from any rate cell must not cross-subsidize or be cross-subsidized by payments for any other rate cell" (U.S. Congress 2016, p. 2). Consequently, CMS is legally obligated to confront cross-subsidization practices in MA. However, owing to the fact that most MA health plans are not mandated to submit their claims data to CMS, the agency often lacks comprehensive insight into MA health plans' operations, which creates a barrier to overseeing this market properly. Therefore, we argue that an increase in transparency in MA claims data are imperative to enable CMS to more effectively administer the MA program. Indeed, the Health Subcommittee of the U.S. House Committee on Energy and Commerce has recently proposed legislation with the aim of enhancing transparency within MA, striving to ensure that policymakers have a lucid comprehension of "how these private plans are utilizing taxpayer dollars" (Committee for a Responsible Federal Budget 2022).

The remainder of this paper proceeds as follows. We review the relevant literature in Section 2 and develop our research hypotheses in Section 3. In Section 4, we present the institutional background for MA with an emphasis on relevant ACA policy changes regarding MA capitation payments. In Section 5, we introduce our data sources and construct the variables used in our econometric models. In Section 6, we develop the econometric model to test our main hypothesis. In Section 7, we empirically assess the impact of MA health plans' strategic cross-subsidization practice on the risk selection problem in MA. We conduct various robustness checks in Section 8 to examine our main findings. We summarize our findings and conclude in Section 9.

2. Literature Review

Capitation payment models have long been among the most debated issues in healthcare payment reforms. On the one hand, healthcare payers (e.g., CMS) have been among the main advocates for them (Rajkumar et al. 2014, James and Poulsen 2016). It is believed that capitation models would improve the efficiency and value of healthcare delivery "by linking the financial incentives for providers to the total cost and/or quality of care they provide" (Centers for Medicare & Medicaid Services 2015). On the other hand, there has been a discrepancy between the expectations and the actual performance of capitation programs in practice. Specifically, several studies noted that MA, the largest capitation program in the United States, has lower accessibility to long-term and postacute care compared with the standard Medicare coverage (Huesch 2010, Meyers et al. 2018, Schwartz et al. 2019, Figueroa et al. 2020). Although this low service accessibility in MA may not translate into worse healthcare outcomes, for example, 30-day readmission rate (Agarwal et al. 2021), evidence exists that high-risk patients have a worse experience of care in MA compared with other Medicare programs (Keenan et al. 2009, Martino et al. 2016, Halpern et al. 2017). Furthermore, when getting sicker, MA patients tend to switch back to TM (Li et al. 2018, Meyers et al. 2019, Ankuda et al. 2020). Healthcare inequality is a main concern of capitation programs as it is in direct contradiction with the purpose of capitation payment models (Rice and Smith 1999).

Past literature mostly attributes the healthcare inequality in capitation programs to the misspecification of risk adjustment formulas. Specifically, Glazer and McGuire (2000) theoretically showed that when risk adjustment is imperfect, healthier patients tend to be overcompensated in MA, whereas sicker patients tend to be undercompensated in MA. As such, MA health plans would have incentives to attract the overcompensated patients and to deter the undercompensated patients, which results in a disparity of service provisions. In fact, Brown et al. (2014) estimated that the annual healthcare costs of each MA enrollee were on average \$317 lower than their capitation rates after risk selection, and further empirically demonstrated that imperfect risk adjustment can lead to risk selection in MA.

Findings in this literature, however, have not been corroborated by subsequent empirical studies on MA. In particular, Newhouse et al. (2013, p. 13) found that MA health plans did not over-provide high-profit healthcare services and under-provide low-profit healthcare services in practice. Therefore, they argued that even if the current MA risk adjustment formula is misspecified and provides incentives for MA health plans to risk select patients, "there is no evidence here that plans act on this incentive." In other words, it is empirically not clear whether MA health plans always act on the incentives to "cherry pick" patients with high profitability.

The presence of strategic cross-subsidization practices helps reconcile the discrepancy between Brown et al.

(2014) and Newhouse et al. (2013) by demonstrating that risk selection is not solely driven by the profitability difference between patients with high-risk and low-risk conditions. Specifically, if MA health plans can cross-subsidize capitation payments across patients, they would "cherry-pick" patients not only according to how profitable these patients are but also how costly it is to attract these patients (She et al. 2022). In fact, because it is easier and less costly for MA health plans to attract low-risk patients compared with high-risk patients (Cooper and Trivedi 2012, Han and Lavetti 2017, Aizawa and Kim 2018, Meyers et al. 2018), MA health plans often reallocate capitation payments to offer benefits that appeal more to low-risk patients, even at the expense of those benefits valued by high-risk patients (Skopec et al. 2019b). Therefore, a widening of profitability differences between high-risk and low-risk patients, where high-risk patients receive deeper capitation payment cuts compared with low-risk patients, may not necessarily worsen risk selection in MA. Our empirical findings in Section 7 provide evidence that supports this implication.

This study also sheds light on recent debates regarding overpayment in MA (Ginsburg and Lieberman 2022). A recent report to Congress delivered by the Medicare Payment Advisory Commission (2022) shows that CMS has consistently overpaid MA plans during the past 20 years compared with the amount TM would have spent on the same beneficiaries. To reduce this overpayment, ACA required CMS to start removing the statutory overpayment in MA from 2012. As a result, the overpayment in MA as a percentage of TM spending has dropped more than 50%, or $68 billion, in the past decade (Weiner 2020, Medicare Payment Advisory Commission 2022). Nevertheless, a recent study shows that MA enrollees' access to or affordability of care was not significantly affected by this ACA payment cut (Skopec et al. 2019a). Our paper further analyzes this policy shock by examining MA plans' changes in spending patterns after the payment reduction and explains why accessibility and affordability of care in MA were not significantly affected. Particularly, we find that deeper capitation payment cuts for high-risk patients unexpectedly lead to more substantial spending reductions for low-risk, rather than high-risk, MA enrollees due to the presence of the strategic cross-subsidization practices.

Our paper has important implications for the operations management literature on healthcare payment reforms. Specifically, the existing literature on population-based payment models shows that failure to accurately estimate the certainty equivalent healthcare costs when determining capitation payments can lead to patient selection (Ata et al. 2013, Adida et al. 2016). Our findings complement and extend this line of research by empirically demonstrating that cross-subsidization incentives can also lead to patient selection in population-based payment models such as capitation. Furthermore, by empirically identifying the cross-subsidization practice, our work helps provide a more complete understanding of healthcare payment models based on past cost estimates (Erhun et al. 2015, Savva et al. 2019) and shows that healthcare costs can be endogenous to healthcare payment models.

Our work also contributes to the empirical operations management literature on cherry-picking behavior in healthcare settings. Patterson et al. (2016) studied patient waiting time in emergency departments and found that resident physicians tended to pick up patients with less complex complaints and avoid patients with more complex complaints. Similarly, Ibanez et al. (2018) empirically showed that physicians preferred to prioritize tasks with shorter expected processing time, albeit it leads to slower completion time on average. In addition, Kc et al. (2020, p. 1) demonstrated that physicians cherry-pick tasks "because of both fatigue and the sense of progress individuals get from task completion." Furthermore, Kc and Terwiesch (2011) provided evidence showing that these cherry-picking behaviors were also found at the hospital level. Recently, an emerging literature in operations management (OM) focuses on analyzing the cherry-picking behavior, or risk selection, at the health plan level (Hwang et al. 2023, Tezcan 2023). In particular, Hwang et al. (2023) analytically demonstrates why MA health plans are incentivized to strategically design their supplementary services to attract profitable Medicare beneficiaries rather than to reduce future treatment costs. Building on this growing body of work, our research presents empirical evidence that suggests strategic cross-subsidization as a potential mechanism through which MA health plans could finance these supplementary services.

Finally, our paper is broadly related to the empirical operations literature that focuses on identifying strategic behaviors in healthcare operations. Bastani et al. (2019) found that hospitals strategically respond to quality improvement regulations by "upcoding" their patients. Chen and Savva (2018) empirically documented that hospitals strategically utilized observation beds to avoid penalties from the Hospital Readmissions Reduction Program. Lu and Lu (2017) found that regulations limiting nurse overtime unintentionally reduce service quality as nursing homes strategically replace permanent nurses with contracted nurses to circumvent regulations. Our study enhances this literature and identifies the cross-subsidization behaviors of MA health plans as a strategic response to the MA capitation payment model.

3. Hypotheses Development

The main hypothesis of this paper aims to investigate whether MA health plans engage in strategic cross-subsidization of capitation payments among different groups of patients. Specifically, when forced to make tradeoffs between various benefits, MA health plans might need to curtail some benefits to "free up funds" for other benefits. Additionally, MA health plans tend to prioritize lower-cost benefits over those targeting high-cost patients (Skopec et al. 2019b). As such, we hypothesize that strategic cross-subsidization is a potential mechanism through which MA health plans finance some of their supplementary or ancillary services, which disproportionately benefit low-risk patients with funds intended for high-risk patients. Notably, the cross-subsidization direction is hypothesized to be from high risk to low risk. This is consistent with the theoretical literature, which suggests that it is unlikely for the opposite direction to be an equilibrium outcome in an individual health insurance market like MA (Azevedo and Gottlieb 2017, Geruso et al. 2023). Formally, the following hypothesis is proposed and tested.

Main Hypothesis (Hypothesis 1). *MA health plans conduct strategic cross-subsidization by reallocating the capitation payments from one group of MA enrollees (e.g., with high expected costs) to another (e.g., with low expected costs).*

It is worth noting that the cost-saving potential of supplemental benefits may offset their expenses. For instance, by investing in fitness membership benefits, MA health plans may assist enrollees in maintaining regular exercise routines, leading to potential long-term cost reductions. However, the lack of transparency in MA claim data grants MA health plans substantial flexibility in resource reallocation (Department of Health and Human Services, Office of Inspector General 2022), presenting multiple avenues to finance benefits. Indeed, our study examines whether and to what extent such benefits may be funded through possibly illegal mechanisms such as strategic cross-subsidization.

To provide a further understanding of the strategic cross-subsidization hypothesis, we conducted a detailed investigation into its effects on risk selection within MA. As outlined in the Introduction, the ACA policy change resulted in more significant capitation payment reductions for high-risk patients compared with their low-risk counterparts. In the presence of cross-subsidization practices, these more substantial capitation payment cuts for high-risk patients would lead to greater spending reductions for low-risk rather than high-risk MA enrollees. Consequently, the ACA policy change would alleviate the risk selection issue within MA. This potential influence of strategic cross-subsidization on MA is encapsulated in the following hypothesis.

Hypothesis 2. *The ACA capitation payment reductions mitigated risk selection in MA.*

Notably, the prevailing risk selection theory, which ignores the cross-subsidization practices (Brown et al. 2014), predicts the opposite outcome. That is, in the absence of cross-subsidization, the reductions in MA capitation payments induced by ACA would exacerbate the risk selection problem in MA, as it further intensifies the underpayment problem for high-risk patients. Therefore, if Hypothesis 2 holds true, it would also substantiate the existence of strategic cross-subsidization in MA.

4. Institutional Background of MA

The first part of this section provides a general introduction to the MA program, with emphasis on its capitation payment model (Section 4.1). The second part discusses the ACA as a shock to this capitation payment model in MA (Section 4.2).

4.1. MA and Its Capitation Payment Model

MA is an insurance program that provides coverage to Medicare beneficiaries, that is, disabled or senior Americans (at least aged 65), for their healthcare services. It was formally established by the Balanced Budget Act of 1997 as an option for Medicare beneficiaries to receive TM benefits through private health plans. More specifically, CMS outsources the inpatient hospital and outpatient services to health plans in MA and reimburses these services through risk-adjusted capitation payments. By 2019, 34% or 22 million Medicare beneficiaries have enrolled in Medicare Advantage, which makes MA the largest capitation program in the United States (Jacobson et al. 2019b).

As a payer in MA, CMS's main responsibility is to use capitation payments to incentivize MA health plans to deliver similar healthcare services to MA enrollees as those in Traditional Medicare. Specifically, the Social Security Act requires CMS to set the MA capitation rates so that the benefits delivered in MA are actuarially equivalent to those in TM. To this aim, CMS first estimates the risk scores ($RiskScore_{i,t}$) of each Medicare beneficiary i based on individual risk factors such as age, gender, and pre-existing conditions at time t using the CMS hierarchical condition categories (CMS-HCC) risk adjustment formula (Centers for Medicare & Medicaid Services 2017). In addition, CMS calculates the benchmark payments ($Benchmark_{c,t}$) for each average (standard risk) enrollee of MA health plans in county c. As such, to provide equivalent healthcare services as those in TM to an MA enrollee i living in county c in year t, the capitation payment an MA health plan receives from CMS is

$$CapitationRate_{i,t} = RiskScore_{i,t} \times Benchmark_{c,t}, \quad (1)$$

which is adjusted yearly.

4.2. ACA Reform of MA Capitation Payments

The ACA, which was signed into law in 2010, required CMS to reduce the benchmark payment ($Benchmark_{c,t}$) in MA capitation rates (1). The main motivation for this reduction was to reduce the overpayment MA health plans received for providing healthcare services to MA beneficiaries before the 2010 ACA reform.

These benchmark payment reductions were not uniform across U.S. counties. Specifically, counties with lower per capita costs of TM compared with the national per capita costs of TM would have larger projected benchmark reductions. Figure 3 in Online Appendix A provides a heatmap to visualize the projected benchmark reductions for each U.S. county. These benchmark payment reductions range from around \$0 per beneficiary per month (PBPM) in some counties to around \$300 PBPM in other counties. In other words, some counties have disproportionately large benchmark payment reductions compared with other counties. The ACA recognized these differential impacts of the benchmark payment reductions on different counties and designed a phase-in scheme to smooth out the transition process, which is explained in detail in Section 6.3.2. This phase-in scheme forms the basis of the identification strategy used in this paper.

5. Data Sources and Variables' Construction

Data used in this study are mainly from two sources. To test the main hypothesis (Hypothesis 1), we use individual patient-level data from Optum's de-identified Clinformatics Data Mart (CDM) between 2009 and 2015. This study period contains years before (2009–2011) and after (2012–2015) the ACA capitation payment cuts in which the same CMS-HCC risk adjustment formula was used in MA. Therefore, this is a suitable time frame for employing our DID identification strategy. To test Hypothesis 2, we use county-level data from the public use files on CMS websites in the same study period. Sections 5.1 and 5.2 introduce these two data sources and explain the construction of relevant variables from these data sets.

5.1. Individual-Level Data

The individual-level data used in this study comes from CDM, a commercial insurance claims database that contains medical claims from more than 150 million people in the United States (Wallace et al. 2014). In this study, we focus on a subsample of this database consisting of Medicare beneficiaries who enrolled in MA for at least one year from 2009 to 2015 and were at least age 65 during the time of enrollment. We excluded approximately 8% of the Medicare beneficiaries in our study sample who moved to another county during the study period. As the healthcare cost estimate of an MA enrollee depends on the county in which this enrollee lived, that is, $\text{Estimated Cost}_{i,t} := RiskScore_{i,t} \times \text{FFS Rate}_{c,t}$, it is not possible to make a reliable estimation of this variable if a patient lived in more than one county in year t. Furthermore, we exclude MA enrollees with risk scores above 7.5, that is, the top 0.1 percentile sickest patient in MA, because these patients typically experienced medical complications that were not representative of the MA population.

The final sample contains MA claims data from 3,882,330 distinct Medicare beneficiaries living in 2,787 counties or county equivalent in 50 states and DC in the U.S. across seven years (2009–2015). The individual-level data set is used to construct variables to test the main hypothesis. These variables are illustrated as follows. $RiskScore_{i,t}$ is the risk score of Medicare beneficiary i in MA at the beginning of year t. We calculate this score using the established CMS-HCC risk adjustment formula, which requires inputs of this Medicare beneficiary's age, gender, and preexisting conditions at the end of year $t-1$ (Centers for Medicare & Medicaid Services 2017). The CDM provides these individual characteristics of Medicare beneficiary i to calculate the beneficiary's risk score. $\text{MA Spending}_{i,t}$ is the total amount an MA health plan spent on an MA beneficiary i in year t. Specifically, we calculate this variable by summing up all MA medical claims generated by patient i and paid by the patient's MA health plan in year t.[1] $\text{FFS Rate}_{c,t}$ is the estimated TM spending on an average (standard risk) MA beneficiary in county c during year t. We calculate this variable using the 2009–2015 county-level TM data publicly available on the CMS's website (Centers for Medicare & Medicaid Services 2019). $\text{Estimated Cost}_{i,t}$ is the estimated amount CMS would spend on patient i in year t if this patient was in TM during year t. Specifically, we calculate this variable as $\text{Estimated Cost}_{i,t} = RiskScore_{i,t} \times \text{FFS Rate}_{c,t}$ based on the capitation payment Equation (1), with the MA benchmark payment $Benchmark_{c,t}$ replaced by the TM benchmark spending $\text{FFS Rate}_{c,t}$. $\text{Age}_{i,t}$ is the age of MA enrollee i in year t. $Treatment_{i,t}$ is a binary variable indicating whether the county in which Medicare beneficiary i lived was in the treatment group (=1) or the control group (=0) in year t. The assignment rule for treatment and control groups is discussed in Section 6.3.2.

In Online Appendix A, we present tables (Tables 6 and 7) summarizing county-level characteristics (e.g., enrollment size and quality star ratings of different MA health plan types), as well as individual-level characteristics (e.g., risk scores and healthcare spending) of CDM throughout our study period (2009–2015).

5.2. County-Level Data

All data used to conduct the county-level analysis in this study are publicly available on the CMS website. Specifically, we use county-level MA data, TM data, and health-plan-level MA data from Centers for Medicare & Medicaid Services (2019). We focus on the analysis of nonspecial needs Health Maintenance Organizations (HMOs) in the main text as they are the dominant health plan type in this market (Song et al. 2013). We also present an analysis for both HMOs and Preferred Provider Organizations (PPOs) in Online Appendix B.5. The final sample in this data set contains observations from 2,968 counties or county equivalent in 50 states and District of Columbia in the United States across seven years (2009–2015).

The county-level data set is used to construct variables to test Hypothesis 2. Specifically, $RiskScore_{c,t}$ is the average risk score of MA beneficiaries in county c of year t. $Treatment_{c,t}$ is a binary variable indicating whether county c was in the treatment group (=1) or the control group (=0) in year t, as discussed in Section 6.3.2. $Star_{c,t}$ is the average star rating of health plans in county c in year t. $McPop_{c,t}$ is the population size of Medicare beneficiaries, including both TM and MA, in county c of year t. $Rebate_{c,t}$ is the average rebate amount for MA health plans to provide supplementary benefits in county c of year t. $RUCC_{c,2010}$ is the Rural-Urban Continuum Code (ranging from 1 to 9), which classifies the rural status of each U.S. county in 2010 (U.S. Department of Agriculture 2013). Here, the higher the $RUCC_{c,2010}$, the more rural county c is. $\text{Income}_{c,2010}$ is the median household income of county c in year 2010.

6. Empirical Evidence of Strategic Cross-Subsidization

This section provides empirical evidence of strategic cross-subsidization in MA. Specifically, Section 6.1 introduces the primary outcome variable of this analysis. Subsequently, Section 6.2 leverages this variable to offer descriptive evidence of existing healthcare disparities in MA. Finally, Section 6.3 empirically identifies strategic cross-subsidization as a cause of these healthcare disparities in MA.

6.1. Spending-Cost Difference

This paper's primary outcome variable, termed *spending-cost difference*, represents the difference between the actual healthcare spending by MA plans on their enrollees ("spending") and the expected healthcare costs for these enrollees as estimated by CMS ("cost"). We calculate this variable by subtracting an MA enrollee's annual healthcare cost estimate from their actual annual healthcare spending for each calendar year in our study period (2009–2015). The *spending* component reflects the actual allocation of healthcare resources by MA plans during a given year, whereas the *cost* component represents the expected amount of annual healthcare spending CMS would have spent on an MA enrollee under TM, adjusted for the enrollee's age, gender, and preexisting conditions (Pope et al. 2011).

The spending-cost difference is a critical metric for analyzing MA health plans' operations. When this measure is positive, indicating spending surpasses the estimated cost for certain MA enrollees, it implies that MA health plans may be overspending on these individuals, or CMS might be underestimating their healthcare costs. Conversely, a negative spending-cost difference, where spending is lower than the estimated costs, points to potential underspending by MA health plans or an overestimation of costs by CMS. This key outcome variable is instrumental in assessing whether there's a systematic disparity between the actual healthcare spending by MA plans and the estimated costs for their enrollees.

6.2. Healthcare Disparities in MA

It is well documented in the literature that spending-cost differences vary among different MA enrollees. On the one hand, it has been documented that MA health plans tend to offer better coverage in terms of preventive care accessibility (e.g., mammographic screening) and supplementary benefits (e.g., gym memberships) that are mainly appealing to healthier Medicare beneficiaries (Cooper and Trivedi 2012, Hung et al. 2016). On the other hand, compared with Medicare beneficiaries in TM, MA enrollees have lower accessibility to high-quality long-term and specialized care, for example, nursing homes (Meyers et al. 2018), home health services (Schwartz et al. 2019), postacute care facilities (Figueroa et al. 2020), and cardiologists (Huesch 2010). Clearly, as an MA patient became sicker, they would utilize less preventive care but more nursing home care or postacute care. This skewed provision of services essentially results in overspending on patients with low risk and underspending on high-risk patients. Consequently, we expect to observe *an inverse relationship between the sickness levels of patients and the spending-cost differences of such patients*, serving as a key descriptive indicator of potential healthcare disparities in MA.

To examine this potential disparity, we analyze the association between Medicare beneficiaries' risk scores ($RiskScore_{i,t}$) and their spending-cost differences (MA $\text{Spending}_{i,t}$ − Estimated $\text{Cost}_{i,t}$). CMS assigns a risk score, $RiskScore_{i,t}$, to each Medicare beneficiary i in MA at the beginning of each year t. Following the existing literature on risk selection (Newhouse et al. 2013, Brown et al. 2014), we use a patient's risk score as a proxy for the sickness level of this patient. We defined the spending-cost difference of an MA health plan enrollee i in year t as the difference between MA health plans' spending on patient i in year t and the estimated healthcare cost of this patient in year t, that is, MA $\text{Spending}_{i,t}$ − Estimated $\text{Cost}_{i,t}$. As detailed in Section 5.1, MA $\text{Spending}_{i,t}$ represents the aggregate expenditure by an MA health plan on its enrollee i in year t, whereas Estimated $\text{Cost}_{i,t}$ is the expected amount

CMS would have expended on patient i in year t had the patient been enrolled in TM during year t. CMS estimated this cost using the CMS-HCC risk adjustment formula, taking into account factors such as the patient's county of residence, age, gender, and pre-existing conditions.

The corresponding econometric model is thus specified as

$$\text{MA Spending}_{i,t} - \text{Estimated Cost}_{i,t} = \beta_0 + \beta_{RS} RiskScore_{i,t} + \alpha_i + \gamma_t + \epsilon_{i,t}, \tag{2}$$

where α_i and γ_t are the individual and time fixed effects, respectively. Individual fixed effects control for the service utilization patterns of different Medicare beneficiaries, whereas the time fixed effect takes into account any pure temporal changes in the study period.

A negative coefficient for β_{RS} would indicate an inverse relationship, where sicker patients (higher risk scores) experience a smaller increase in MA spending compared with the increase in their estimated healthcare costs. This would be consistent with potential under-spending on higher-risk patients, a key concern regarding healthcare disparities in MA.

We estimate the model using two-way fixed effects and present the results in Table 1. The estimated coefficient for β_{RS} is negative and statistically significant, confirming the expected inverse relationship. This implies that a one-unit increase in a patient's risk score is associated with a more than \$9,000 decrease in the relative annual MA health plan spending on this patient compared with the estimated healthcare cost, that is, spending-cost difference. This finding suggests that sicker patients experience a larger discrepancy between their actual spending and their expected cost, indicating potential healthcare disparities within the MA program.

6.3. Main Hypothesis (Hypothesis 1)

This section aims to identify strategic cross-subsidization as a cause of the inverse relationship between the sickness of MA beneficiaries and their spending-cost differences. Specifically, the inverse relationship can be explained by two plausible mechanisms. (1) MA health plans' strategic reallocation of capitation payments from one group of MA enrollees (e.g., those with high expected costs) to another (e.g., those with low expected costs), as hypothesized in our main hypothesis (Hypothesis 1). (2) CMS's overestimation of the cost for low-risk patients and the underestimation of the cost for high-risk patients. In what follows, by making use of an external shock (ACA payment cuts), we provide strong evidence for the first mechanism. Specifically, our external shock shifts spending on MA patients (MA Spending$_{i,t}$) without affecting how their healthcare costs (Estimated Cost$_{i,t}$) were estimated, thereby ruling out the alternative overestimation explanation.

Section 6.3.1 discusses that the ACA capitation rate cuts provided such an exogenous shock and thus can serve as our source of identification. Section 6.3.2 discusses the assignment of treatment and control groups in this quasi-natural experiment. Finally, we develop the econometric model in Section 6.3.3 and present the estimation results in Section 6.3.4.

6.3.1. Identifying Strategic Cross-Subsidization Through ACA Capitation Rate Cuts. As discussed in Section 4.2, the ACA reform changed MA capitation payments by reducing their benchmark payments. Specifically, we note that the ACA capitation rate cuts have an asymmetric impact on MA patients in different risk groups in that patients with higher risk scores experienced deeper cuts in their capitation payments ($CapitationRate_{i,t}$). The reason is that the ACA capitation rate cuts only affect $CapitationRate_{i,t} := RiskScore_{i,t} \times Benchmark_{c,t}$ (c.f. (1)) through the benchmark payments ($Benchmark_{c,t}$). As such, the same benchmark payment cuts would result in deeper capitation payment cuts for MA patients with higher risk scores ($RiskScore_{i,t}$). Therefore, the ACA capitation rate cuts essentially exogenously reduce a greater amount of capitation payments from high-risk patients than from low-risk patients. Notably, the ACA cuts do not affect the healthcare cost estimates of MA enrollees, which are defined as Estimated Cost$_{i,t} := RiskScore_{i,t} \times$ FFS Rate$_{c,t}$ in Section 5.1.

Based on these observations, we can empirically identify the strategic cross-subsidization practice of MA health plans by examining how the observed inverse relationship between risk scores and spending-cost differences would change after the ACA shock. As illustrated in Figure 1(a), if there were no strategic cross-subsidization, the ACA shock, with its deeper payment cut for high-risk patients, would simply amplify the inverse relationship between risk scores and spending-cost differences. In contrast, in the presence of strategic cross-subsidization, a deeper payment cut for high-risk patients can result in a deeper spending cut for low-risk patients, as shown in Figure 1(b).

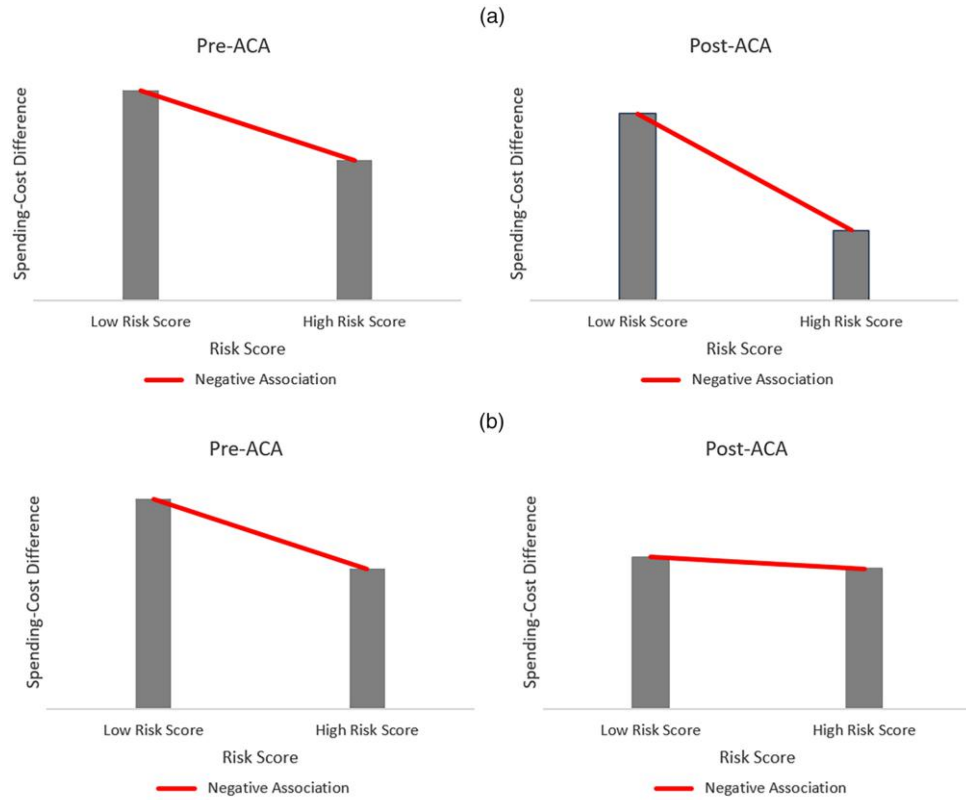
Table 1. Estimation Results of Econometric Model (2)

Model parameters	Spending-cost difference
β_{RS}	−9.97***
Observations	12,937,959
R^2	0.533
Adjusted R^2	0.330

Notes. Dependent variable is in \$1,000 per year unit. Association between MA enrollees' spending-cost differences and their risk scores. All standard errors are clustered at county level.

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

Figure 1. (Color online) Identification Strategy



Notes. (a) Identification strategy based on ACA cut (no cross-subsidization): In the absence of strategic cross-subsidization, the ACA shock would not mitigate the inverse relationship between risk scores and spending-cost differences. Here, the inverse relationships in the Pre-ACA and Post-ACA figures correspond to β_{RS} and $\beta_{RS} + \delta_{DID}$ in (3), respectively. (b) Identification strategy based on ACA cut (cross-subsidization): In the presence of strategic cross-subsidization, the ACA shock would mitigate the inverse relationship between risk scores and spending-cost differences. Here, the inverse relationships in the Pre-ACA and Post-ACA figures correspond to β_{RS} and $\beta_{RS} + \delta_{DID}$ in (3), respectively.

Hence, to test the main hypothesis (i.e., Hypothesis 1), it suffices to check whether the inverse relationship between risk scores and spending-cost differences was alleviated by the ACA shock. If further underpayment for high-risk patients indeed results in amplified underspending on these patients, we would anticipate an intensified inverse relationship following the ACA shock, as illustrated in Figure 1(a). Alternatively, if we observe a pattern akin to Figure 1(b), it suggests that exacerbated underpayment for high-risk patients does not instigate further underspending on these patients, but instead mitigates overspending on their low-risk counterparts. Consequently, the heightened spending-cost differences among low-risk patients and diminished spending-cost differences among high-risk patients cannot be solely ascribed to overpayment for the former and underpayment for the latter. A cross-subsidization mechanism, therefore, becomes essential to explain the inverse relationship between risk scores and spending-cost differences.

6.3.2. Treatment Assignment. To effectively isolate the impact of ACA capitation rate cuts on the inverse relationship between risk scores and spending-cost differences from other confounding factors, we need a near-random assignment of capitation rate cuts across U.S. counties. In this section, we first illustrate the staggered roll-out of ACA capitation rate cuts across the United States introduced in Section 4.2 and subsequently explain how this staggered roll-out provides a quasi-natural experiment to establish treatment and control groups.

To smooth the transition from pre-ACA to ACA benchmark payments, ACA specified a six-year transition period (2012–2017) to gradually phase in these payment reductions. Specifically, ACA divided counties across the United States into three groups, namely the two-, four-, and six-year phase-in groups. In the transition period (2012–2017), ACA benchmark payments were phased into different phase-in groups in a staggered manner, according to the phase-in factor ($PhaseInFactor_{c,t}$) listed in Table 2. More precisely, the benchmark payments for each MA health plan in county c during years $t \in \{2012, \ldots, 2017\}$ were calculated as $Benchmark_{c,t} = ACA\_Benchmark_{c,t} \times PhaseInFactor_{c,t} + PreACA\_Benchmark_{c,t} \times (1 - PhaseInFactor_{c,t})$, where

Table 2. Phase-in Factor of ACA Capitation Rates of County c in Year t($PhaseInFactor_{c,t}$)

	Two-year phase-in group		Four-year phase-in group		Six-year phase-in group	
Year	Pre-ACA	ACA	Pre-ACA	ACA	Pre-ACA	ACA
2012	1/2	1/2	3/4	1/4	5/6	1/6
2013	0	1	1/2	1/2	2/3	1/3
2014	0	1	1/4	3/4	1/2	1/2
2015	0	1	0	1	1/3	2/3
2016	0	1	0	1	1/6	5/6
2017	0	1	0	1	0	1

Notes. Counties across the United States phased into the ACA benchmark in three groups. Two-year phase-in group, these counties phased in $\frac{1}{2}$ of the ACA capitation rates in 2012 and fully phased in afterward; four year phase-in group, these counties phased in $\frac{1}{4}$ of the ACA capitation rates in 2012, phased in $\frac{1}{2}$ of the ACA capitation rates in 2013, phased in $\frac{3}{4}$ of the ACA capitation rates in 2014, and fully phased in afterward; Six-year phase-in group, these counties phased in $\frac{1}{6}$ of the ACA capitation rates in 2012, phased in $\frac{1}{3}$ of the ACA capitation rates in 2013, phased in $\frac{1}{2}$ of the ACA capitation rates in 2014, phased in $\frac{2}{3}$ of the ACA capitation rates in 2015, phased in $\frac{5}{6}$ of the ACA capitation rates in 2016, and fully phased in afterward (Centers for Medicare & Medicaid Services 2011).

$PhaseInFactor_{c,t} \in [0,1]$ as specified in Table 2. In other words, the actual MA benchmark payments in the transition period were a weighted average of the ACA benchmark payments ($ACA\_Benchmark_{c,t}$) and the pre-ACA benchmark payments ($PreACA\_Benchmark_{c,t}$), where counties c with higher "weight" ($PhaseInFactor_{c,t}$) would phase into the ACA benchmark earlier.

To see how this phase-in group assignment smooths the transition from the pre-ACA benchmark to the ACA benchmark, we shall explain how the $PhaseInFactor_{c,t}$ presented in Table 2 was determined. Specifically, $PhaseInFactor_{c,t}$ was calculated based on its projected difference between the ACA and Pre-ACA benchmark payments, that is, $PreACA\_Benchmark_{c,t} - ACA\_\ Benchmark_{c,t}$. The idea is that counties with larger projected benchmark cuts would have lower phase-in factors ($PhaseInFactor_{c,t}$) and thus longer phase-in periods. More precisely, counties with projected benchmark payment cuts less than \$30 PBPM were assigned to the two-year phase-in group, whereas counties with at least \$30 PBPM projected benchmark cuts were assigned to groups with longer, that is, four- or six-year, phase-in periods. This phase-in group assignment rule ensures that no counties would experience disproportionately large annual benchmark cuts during the transition period and thus smooths the transition process. For example, a county with \$40 PBPM projected benchmark cut would be assigned to the four-year phase-in group, and thus only had $\$40 \times \frac{1}{4} = \$10$ PBPM benchmark cut in 2012. In contrast, a county with \$20 PBPM projected benchmark cut would also have $\$20 \times \frac{1}{2} = \$10$ PBPM benchmark cut in 2012, because it is in the two-year phase-in group. In Section 6.3.2, we describe in detail how we exploit this phase-in group assignment rule to implement our DID design.

As such, each U.S. county was assigned to different phase-in groups in our study period. Furthermore, these group assignments were based on the projected difference between the pre-ACA benchmarks and ACA benchmarks of each county. These two institutional details provide us with the quasi-random regional variations of ACA capitation rate cuts for treatment assignment.

First, the phase-in group assignment of each U.S. county (c.f. Table 2) enables us to implement the DID design. Specifically, during the transition period, counties with shorter phase-in time had higher percentage capitation rate cuts compared with counties with longer phase-in time. For example, 50% of the ACA capitation payment cuts became effective in 2012 for counties in the two-year phase-in group. In contrast, for counties in the four-year and six-year phase-in groups, only 25% and 17% of the ACA capitation payment cuts became effective in 2012. Therefore, in principle, counties with shorter phase-in time should be assigned to the treatment group, while counties with longer phase-in time should be assigned to the control group.

Second, phase-in time for counties was determined based on their projected difference between the pre-ACA benchmarks and ACA benchmarks and whether this value fell above or below an arbitrary cut-off point. In particular, counties with projected benchmark differences below \$30 PBPM were assigned to the two-year phase-in group, whereas counties with projected benchmark differences of at least \$30 PBPM were assigned to the four-year and six-year phase-in groups. As such, this \$30 PBPM provides a clean cut-off value to implement a "local randomized experiment" as defined by Lee and Lemieux (2010). Specifically, for counties with projected benchmark difference slightly below \$30 PBPM, they experienced $50\% - 25\% = 25\%$ deeper capitation payment cuts than counties with projected benchmark difference slightly above \$30 PBPM. Therefore, by focusing on the counties with benchmark differences just above or just below the \$30 cut-off value, we have a study sample where the ACA capitation payment cuts were assigned independent of other temporal changes. As such, we use this \$30 cut-off value to create a quasi-random assignment of the ACA capitation payment cuts across otherwise similar U.S. counties.

In our DID analysis, counties with projected benchmark differences slightly below the $30 cut-off value are assigned to the treatment group, while counties with projected benchmark differences slightly above the $30 cut-off value are assigned to the control group. Specifically, we test our main hypothesis in three different neighborhood specifications around the $30 cut-off value (i.e., [$29,$31], [$28,$32], [$27,$33]), and assess whether they provide consistent treatment effect estimates. Lastly, we remark that the phase-in group assignment and phase-in time were announced in February 2011 based on the projected benchmark difference from the 2010 data (Centers for Medicare & Medicaid Services 2011). Therefore, the treatment path was predetermined before 2012.

6.3.3. Econometric Model to Test the Main Hypothesis.

The econometric model to test our main hypothesis is specified as

$$\begin{aligned} &\text{MA Spending}_{i,t} - \text{Estimated Cost}_{i,t} \\ &= \beta_0 + \delta_{DID} RiskScore_{i,t} \times Treatment_{i,t} + \beta_1 Treatment_{i,t} \\ &\quad + \beta_{RS} RiskScore_{i,t} + \alpha_i + \gamma_t + \epsilon_{i,t}, \end{aligned} \tag{3}$$

where the treatment $Treatment_{i,t}$ is assigned as described in Section 6.3.2. In particular, because treatment and control groups are assigned at the county level, the individual identifiers i become county-patient specific. In this specification, if our main hypothesis was true, then $\delta_{DID} > 0$. The reason is, as explained in Section 6.3.1, that the cross-subsidization mechanism would translate the capitation payment cut for high-risk patients into a spending cut for low-risk patients. As such, the spending-cost difference for low-risk patients would experience a more significant reduction compared with the spending-cost difference for high-risk patients, which would be captured by a mitigated negative slope, as illustrated in Figure 1(b). As such, the inverse relationship between risk scores and spending-cost differences (β_{RS}) estimated in (2) would be mitigated by the ACA capitation rate cuts, that is, $\beta_{RS} < \beta_{RS} + \delta_{DID}$, in the presence of cross-subsidization.

Consistent with prior research employing similar identification strategies (Verhoogen 2008, Irarrazabal et al. 2015, Flach 2016), the econometric model (3) can be systematically formulated in three steps. First, we identify the inverse relationship between risk scores and spending-cost differences in the control, that is, $\beta_{RS(Control)}$. That is, for patients i in the control counties or in the control period (t before the ACA shock), we estimate

$$\begin{aligned} &\text{MA Spending}_{i,t} - \text{Estimated Cost}_{i,t} \\ &= \beta_{0(Control)} + \beta_{RS(Control)} RiskScore_{i,t} + \alpha_i + \gamma_t + \epsilon_{i,t}. \end{aligned}$$

Second, we identify the inverse relationship between risk scores and spending-cost differences in the treatment, that is, $\beta_{RS(Treatment)}$. That is, for patients i in the treatment counties and in the treatment period (t after the ACA shock), we estimate

$$\begin{aligned} &\text{MA Spending}_{i,t} - \text{Estimated Cost}_{i,t} \\ &= \beta_{0(Treatment)} + \beta_{RS(Treatment)} RiskScore_{i,t} + \alpha_i + \gamma_t + \epsilon_{i,t}. \end{aligned}$$

Third, we identify the change in the inverse relationships between control and treatment, that is,

$$\delta_{DID} := \beta_{RS(Treatment)} - \beta_{RS(Control)}. \tag{4}$$

Mathematically, it is clear that the parameter of interest in (3), that is, δ_{DID}, can be identified as (4). Econometrically, Verhoogen (2008, p. 510) noted that this identification strategy "would amount to a familiar triple-differences strategy" if $RiskScore_{i,t}$ was a binary variable. As such, the econometric model (3) can be structured as a triple-difference design with continuous heterogeneous treatment effects, as summarized in Table 5 of Online Appendix A.

6.3.4. Results for the Main Hypothesis.

The econometric model (3) is estimated using two-way fixed effect estimators. Table 3 presents the estimation results from different neighborhood specifications around the $30 cut-off value, that is, [$29,$31], [$28,$32], [$27,$33]. These different neighborhood specifications provide consistent results supporting our main hypothesis, that is, $\delta_{DID} > 0$. In other words, the inverse relationship between the sickness levels of MA beneficiaries and their spending-cost differences was mitigated by the ACA capitation rate cuts, that is, $|\beta_{RS}| > |\beta_{RS} + \delta_{DID}|$. As such, we confirm the main hypothesis and empirically identify the strategic cross-subsidization practice in MA.

Next, we discuss the effect size of the ACA capitation rate cuts on MA health plans' spending patterns. Specifically, the strategic cross-subsidization amount was decreased by around $\delta_{DID} = \$1,640$ (according to the [$28,$32] neighborhood) per patient per year for a one-

Table 3. Estimation Results of Econometric Model (3)

	Spending-cost difference		
Model parameters	[$29,$31]	[$28,$32]	[$27,$33]
β_{RS}	−10.15***	−9.74***	−9.65***
δ_{DID}	2.02***	1.64***	1.51***
β_1	−1.88**	−1.98***	−1.92***
Observations	588,946	779,436	984,339
R^2	0.500	0.508	0.528
Adjusted R^2	0.311	0.309	0.323

Notes. Dependent variable is in $1,000 per year unit. Impact of the ACA cuts on MA enrollees' spending-cost differences. All standard errors are clustered at county level.

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

unit risk score increase after the ACA capitation rate cuts. In other words, the ACA shock mitigated around 17% of the strategic cross-subsidization amount, as measured by $\left|\frac{\delta_{DID}}{\beta_{RS}}\right|$. We note that this effect size is economically significant when taking into account the scale of the MA population. For example, there were around 15 million MA enrollees in 2014 (Jacobson et al. 2016), where the mean annual change in individual risk score was 0.14 (c.f. Table 6 in the Online Appendix). Therefore, the mitigated strategic cross-subsidization in 2014 was around $0.14 \times \$1,640 = \$229.6$ per patient or $15\,\text{million} \times 0.14 \times \$1,640 = \$3,444$ million $\approx \$3$ billion in total. Alternatively, we also make use of quartile statistics of the individual risk score change in 2014 to provide a lower bound for this estimate. Specifically, 25% of our studied population had at least a 0.3 unit increase in risk score during 2014. In this case, the mitigated strategic cross-subsidization in 2014 was at least $0.25 \times 0.3 \times \$1,640 = \$123$ per patient or $15\,\text{million} \times 0.25 \times 0.3 \times \$1,640 = \$1,845$ million $\approx \$2$ billion in total.

7. Assessing the Impact of Strategic Cross-Subsidization Practice on Risk Selection

In the presence of strategic cross-subsidization, Hypothesis 2 posits that the ACA policy change, which imposed more substantial capitation payment reductions for high-risk patients, would mitigate risk selection in MA, as detailed in Section 3. To empirically assess this hypothesis, we rely on county-level data as described in Section 5.2 rather than individual-level data from CDM. The reason is that OPTUM does not publish the design weights, that is, the inverse of sampling probability, for users to recover population estimates. Section 7.1 develops a generalized triple-difference model to examine this association. Section 7.2 presents and discusses the estimation results.

7.1. Econometric Model to Test Hypothesis 2

One way to evaluate Hypothesis 2 is to analyze whether counties phased into the ACA benchmarks earlier had higher county average MA risk scores in comparison with other counties. Here, a higher average risk score indicates a higher-risk enrolled population, suggesting a mitigated risk selection or cherry-picking behaviors in MA. In other words, the ACA capitation payment reductions should alleviate risk selection in a staggered manner. This means that counties that phased in earlier are expected to show a trend of enrolling a higher risk population.

However, because the ACA shock affects risk selection through multiple channels, it is difficult to isolate the impact of strategic cross-subsidization practice on risk selection in this shock. For example, along with the capitation payment cuts, the ACA also created a quality bonus payments (QBPs) program to incentivize better quality provision in MA through bonus payments (Centers for Medicare & Medicaid Services 2020). It is shown that the QBPs program enabled high-quality health plans to select healthier Medicare beneficiaries (Fioretti and Wang 2023). As such, MA health plans with more quality bonus payments might have more severe risk selection problems. Therefore, the existence of alternative channels through which the ACA shocks can affect risk selection might confound the estimation of the association between strategic cross-subsidization and risk selection.

To control for these alternative channels, we use a generalized triple-difference model to isolate the impact of strategic cross-subsidization on risk selection, that is,

$$\begin{aligned} RiskScore_{c,t} = {} & \beta_0 + \delta_{DID} Treatment_{c,t} + \beta_{Star(DID)} Treatment_{c,t} \\ & \times Star_{c,12} + \beta_{Star(Post)} Star_{c,12} \times I\{PostACA\} \\ & + \beta_{Star} Star_{c,t} + \beta_{McPop} log(McPop_{c,t}) \\ & + \beta_{Rebate} Rebate_{c,t} + \alpha_c + \gamma_t + \epsilon_{c,t}. \end{aligned} \tag{5}$$

That is, in addition to assigning counties to the treatment/control groups, we also group counties by the county-average initial star ratings at the beginning of the QBPs program, that is, $Star_{c,12}$, which determines the county-average additional quality bonus payments from ACA. Notably, these initial star ratings in 2012 were mainly determined by MA health plans' performance in the year 2010, and thus are not affected by the phase-in group assignment published in 2011 (Centers for Medicare & Medicaid Services 2011). As such, the treatment effect is moderated by the additional quality bonus payments from ACA. In addition, we also control for the population size of Medicare beneficiaries, that is, $McPop_{c,t}$, and the county-average funds available for MA health plans to provide supplementary benefits, that is, $Rebate_{c,t}$. Both factors affect the flexibility for MA health plans to conduct strategic cross-subsidization, as well as risk selection.

7.2. Results for Hypothesis 2

Table 4 displays the estimation results of the econometric model (5) for HMO plans. The results, estimated using two-way fixed effect estimators, confirm Hypothesis 2. Specifically, the parameter of interest, δ_{DID}, is statistically significant and positive in all three neighborhood specifications. This finding indicates that health plans in the treatment group enrolled a higher proportion of high-risk Medicare beneficiaries relative to those in the control group counties. Consequently, this pattern implies that health plans in the treatment group are less involved in risk selection practices than their counterparts in the control group. In this context,

Table 4. Estimation Results of Econometric Model (5)

	Risk score		
Model parameters	[\$29,\$31]	[\$28,\$32]	[\$27,\$33]
δ_{DID}	0.185**	0.139**	0.109**
$\beta_{Star(DID)}$	−0.050**	−0.045***	−0.036***
$\beta_{Star(Post)}$	−0.028**	−0.014	−0.020**
β_{Star}	−0.035***	−0.009	−0.008
β_{McPop}	−0.011*	−0.011**	−0.011***
β_{Rebate}	2×10^{-5}	1×10^{-5}	2×10^{-5}**
Observations	226	384	580
R^2	0.860	0.870	0.860
Adjusted R^2	0.825	0.841	0.831

Note. Impact of the ACA cuts on county-average risk scores of MA enrollees in HMO plans.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

risk selection refers to the tendency of health plans to preferentially enroll lower-risk individuals, and a reduction in this practice suggests a more equitable and inclusive approach to beneficiary enrollment. Furthermore, the treatment effect is moderated by QBP star ratings, with $\beta_{Star(DID)} < 0$. Additional results for both HMO and PPO plans are presented in Online Appendix B.5.

The estimation results of the other parameters also have the expected signs. Specifically, we find that MA health plans with higher star ratings tend to have worse risk selection, $\beta_{Star} < 0$. Besides, risk selection is more pronounced in counties with a higher percentage of Medicare-eligible populations, that is, $\beta_{McPop} < 0$. The reason is that MA plans have more flexibility in conducting risk selection in these counties. Lastly, the amount of rebate available for MA health plans to provide supplementary benefits did not significantly impact risk selection during our study period, that is, $\beta_{Rebate} \approx 0$. One reason is that the rebate amount in MA had stayed relatively the same throughout our study period (Guterman et al. 2018).

As for the overall impact of the ACA shock on risk selection, we find that risk selection was mitigated in counties with low initial star ratings but worsened in counties with high initial star ratings. Specifically, for counties with low initial star ratings (e.g., $Star_{c,12} = 3$), risk scores in the treatment group increase after the ACA shock, that is, $\delta_{DID} + \beta_{Star(DID)} Star_{c,12} > 0$. However, for counties with high initial star ratings (e.g., $Star_{c,12} = 4$), risk scores in the treatment group may even decrease after the ACA shock, that is, $\delta_{DID} + \beta_{Star(DID)} Star_{c,12} < 0$. A possible explanation is that the additional capitation payments from the QBPs program after the ACA shock may help MA health plans conduct more risk selection. Therefore, for these high-quality plans, the negative impact of ACA on risk scores through the channel of QBPs outweighs the positive impact of ACA on risk scores through the channel of reduced strategic cross-subsidization. These results highlight the necessity to examine the heterogeneous treatment effect, instead of the population average treatment effect, when studying the impact of the ACA shock on risk selection in MA.

8. Robustness Checks

This section checks the robustness of our findings. Specifically, Section 8.1 conducts a matching analysis. Section 8.2 examines the parallel trend assumption of our DID analysis. Additional robustness checks are presented in Online Appendix B.

8.1. Matching Analysis

To check whether the discontinuity in ACA capitation cuts discussed in Section 6.3.2 indeed provides such randomization, this section conducts a coarsened exact matching (CEM) on relevant county- and individual-level characteristics and re-estimate our main econometric model (3) using the matched sample.

First, we conduct a CEM using relevant county- and individual-level characteristics introduced in Section 5 and present the results in Tables 8 and 9 of Online Appendix A. Specifically, FFS $\text{Rate}_{c,2010}$, $RUCC_{c,2010}$, and $\text{Income}_{c,2010}$ may affect treatment assignments through MA benchmark payments. Here we focus on 2010 because the treatment and control groups were assigned based on the county-level benchmark difference in 2010. $\text{Star}_{c,2012}$ measures the balance between treatment and control groups in health plan quality and QBPs in 2012. We focus on the year 2012 because the star ratings in 2012 were mainly determined based on MA health plans' performance in 2010, the last year before the treatment and control assignments were announced. $\text{Age}_{i,t}$ is a proxy for the balance of the patient population's health in our study period. In addition, we re-estimate our main econometric model using the matched data and present the estimation results (Table 10 of Online Appendix A). The results based on matched data are consistent with our main results (Table 3).

8.2. Parallel Trend Assumption

A key assumption of our DID empirical strategy is that counties in treatment and control groups were otherwise similar except for their ACA capitation payment cuts. In other words, in a counterfactual world where the ACA capitation payment cuts were not introduced, counties in treatment and control groups would follow the same "parallel trends" so that the treatment effect estimate would not be statistically different between the treatment and control groups. In this section tests, we test this assumption using an event-study design, that is,

$$\begin{aligned}
&\text{MA Spending}_{i,t} - \text{Estimated Cost}_{i,t} \\
&= \beta_0 + \sum_{s=2009}^{2010} (\delta_{DID(s)} RiskScore_{i,t} \times Treatment_{i,s-2011} \\
&\quad + \beta_s Treatment_{i,s-2011}) + \sum_{s=2012}^{2015} (\delta_{DID(s)} RiskScore_{i,t} \\
&\quad \times Treatment_{i,s-2011} + \beta_s Treatment_{i,s-2011}) \\
&\quad + \beta_{RS} RiskScore_{i,t} + \alpha_i + \gamma_t + \epsilon_{i,t}. \qquad (6)
\end{aligned}$$

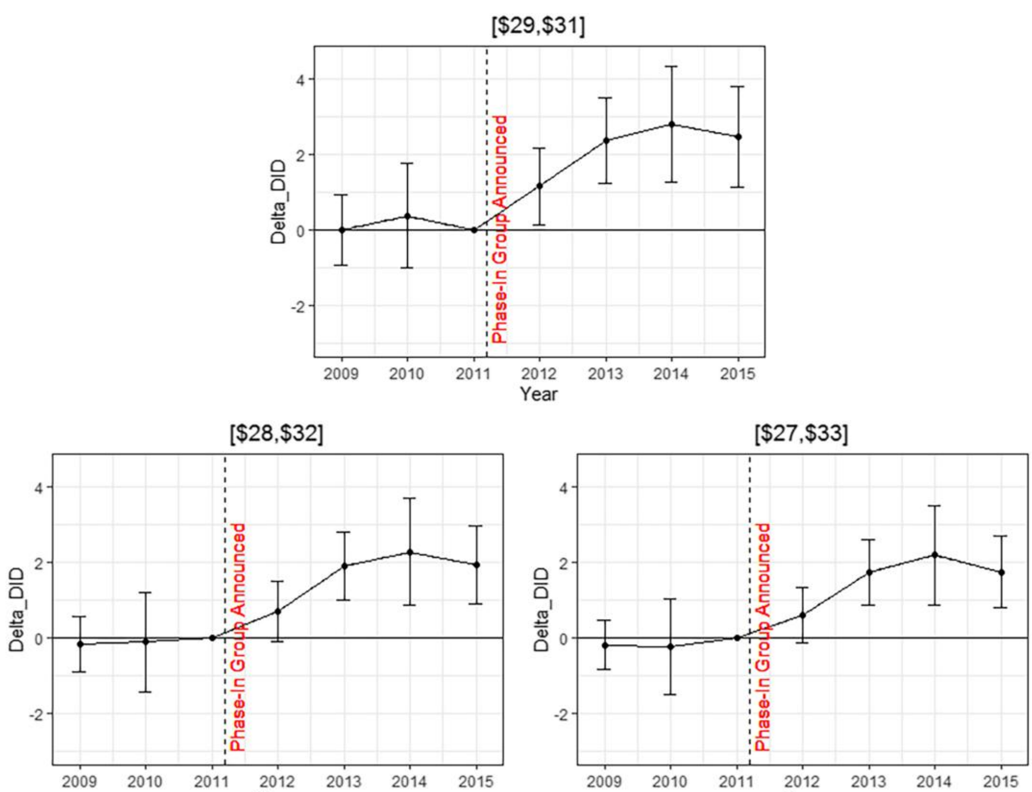
Specifically, if the parallel trend assumption holds, the parameter of interest $\delta_{DID(s)}$ should stay close to zero in the pretreatment period (s = 2009, 2010), which implies that the degree of strategic cross-subsidization was similar between treatment and control groups before the ACA cut. Notably, we set 2011 as the base year because the phase-in group assignment was announced in February 2011 (Centers for Medicare & Medicaid Services 2011).

Estimation results of the event-study model (6) are presented in Table 11 of Online Appendix A. Consistent with the identification assumption, we do not find a statistically significant difference in strategic cross-subsidization practice between the treatment and control groups before the implementation of ACA cuts, that is, $\delta_{DID(2009)} = \delta_{DID(2009)} = 0$. That is, MA health plans in both treatment and control groups have followed parallel trends in terms of their strategic cross-subsidization practice. Their trends started diverging when the staggered rollout of ACA cuts began in 2012, that is, $0 < \delta_{DID(2012)} < \delta_{DID(2013)} < \delta_{DID(2014)}$. When the staggered rollout was completed in 2015 for the local neighborhood studied in this paper, we find that the trends of treatment and control groups started converging again, that is, $\delta_{DID(2015)} < \delta_{DID(2014)}$. We can also visually verify this implication of the parallel trend assumption through an event study plot of $\delta_{DID(s)}$ as in Figure 2.

9. Conclusion

Healthcare capitation payment models, particularly the Medicare capitation program, have been extensively studied in recent decades. However, because of the limited access to MA health plan claims data, little is known about how MA health plans allocate capitation payments across various patient risk groups. More importantly, there is a limited understanding of whether and to what extent there may be perverse incentives at play in MA health plans' operations leading to a systematic mismatch between a patient's health status and the

Figure 2. (Color online) Event Study Plots of $\delta_{DID(s)}$ in Econometric Model (6): Impact of the ACA Cuts on MA Enrollees' Spending-Cost Differences



healthcare resources she gets from MA. Our paper made use of a large commercial insurance database containing medical claims from more than 2 million MA enrollees to open up the black box of MA health plans' operations and to study the allocation problem of MA capitation payments. Our findings shed new light on how MA health plans operate in serving different patient risk groups and how their operations can affect the risk selection problem in MA.

By scrutinizing the capitation payment allocations within MA health plans, this study not only highlights the existence of cross-subsidization but also emphasizes its far-reaching consequences on risk selection within the MA market. Irrespective of the direction in which cross-subsidization occurs, whether from high-risk patients to low-risk patients or vice versa, it has profound implications. In cases where MA health plans utilize capitation payments from high-risk patients to subsidize low-risk patients, a typical risk selection problem emerges. This scenario incentivizes only low-risk patients to enroll in MA plans, leaving sicker patients predominantly to TM. Such a trend can lead to a phenomenon known as market unraveling. In this context, market unraveling refers to the possibility that the CMS may find it increasingly challenging to sustain TM if it becomes disproportionately populated by the sickest beneficiaries of Medicare. This imbalance could pose significant financial and operational challenges for CMS in maintaining TM as a viable option (Medicare Payment Advisory Commission 2023). Conversely, if MA health plans resort to using capitation payments from low-risk patients to subsidize high-risk patients, the issue of adverse selection comes into play. This situation could favor plans targeting low-risk patients, once again resulting in risk selection (Azevedo and Gottlieb 2017, Geruso et al. 2023).

Our findings have direct practical implications. First, it underscores the urgent need for Medicare to enhance its requirements for data reporting and promote improved data transparency when it comes to managing Medicare Advantage plans. Particularly, our study offers compelling rationale and empirical evidence in favor of federal laws pertaining to cross-subsidization practices within the MA landscape, as outlined in the Code of Federal Regulations: "Payments from any rate cell must not cross-subsidize or be cross-subsidized by payments for any other rate cell" (U.S. Congress 2016). Second, regulations on cross-subsidization practices must be upheld to address the potential risk selection problem in MA, where the low-risk population benefits at the expense of the more vulnerable high-risk population. Critically, given the correlation between healthcare risk factors and socioeconomic status, the strategic cross-subsidization tactics adopted by MA plans could worsen socio-economic inequalities within the system, potentially leading to a reallocation of capitation payments away from historically disadvantaged groups (Irvin et al. 2020, Tran 2020, McWilliams et al. 2023).

Our study emphasizes the need for further research in this area. Firstly, with the growing adoption of population-based payment models by healthcare payers, it is likely that strategic cross-subsidization practices extend beyond the MA capitation program. Additional investigations are essential to comprehend the role of strategic cross-subsidization practices in other population-based payment models. Secondly, because of the absence of linked clinical data, we are unable to analyze the effects of strategic cross-subsidization on patient outcomes. We anticipate that our initial exploration of strategic cross-subsidization will catalyze future research aimed at assessing its implications for health outcomes.

Endnote

[1] Optum does not provide the actual price paid by MA health plans for each service and only has the amount charged by physicians and hospitals for each service. Nevertheless, it offers a standardized algorithm to convert these charged prices to actual paid prices. For example, the professional service (e.g., physician visits, surgery, laboratory tests, imaging) charges are standardized using the resource-based relative value scale (RBRVS) approach, a method applied by CMS to price services provided in traditional Medicare. We use these standardized charged prices to create a consistent proxy for the actual expense.

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